

Sablefish Background, Biology, and Key Uncertainties

2026 CIE Review

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Contributors: Kevin Siwicke and Katy Echave

June 2026



Agenda

Tuesday, June 16, 2026

- A. Welcome and Introductions (30min)
- B. Sablefish Background (Management, Biology, and Data Inputs; 3hrs)
- C. General Assessment Model Structure (ToR 1; 3hrs)

Wednesday, June 17, 2026

- A. Reconvene and Review (30min)
- B. Summary of Closed Loop Simulations (3hrs)
- C. Comparing Assessment Performance and Diagnostics (ToRs 1/2; 3hrs)

Thursday, June 18, 2026

- A. Reconvene and Review (30min)
- B. Comparing Assessment Performance and Diagnostics *Cont'd* (ToR 1/2; 3hrs)
- C. Spatial Model Configurations and Diagnostics (ToR 3; 3hrs)

ADJOURN



Terms of Reference (ToRs)

Days 1-2

- 1) Review the disseminated documents and analyses, then provide feedback on whether the fleets-as-areas (FAA) assessment provides a suitable basis for operational management advice, particularly in comparison to the current single region panmictic assessment.
 - a) Provide recommendations on the most appropriate parametrization of fleet structure (number of fleets per fishery/survey, as a proxy for regions) in the FAA model.
- 2) Highlight potential strengths and weaknesses of the FAA assessment, including model assumptions, estimates, fits to data (survey indices and age compositions along with fishery catch-at-age data), model diagnostics, and major sources of uncertainty.
 - a) Indicate whether the model provides the best scientific information available for estimating current stock status, projecting future abundance, and determining Overfishing Levels (OFLs) and Acceptable Biological Catches (ABCs).

Day 3

- 3) Provide research recommendations for improving the existing research-oriented sablefish spatial (i.e., 3 and 5 region) assessment models, especially in the context of potential future usage in a management context.

Reviewers must provide details for the basis of their responses to the review ToRs.



Documents and Materials

- CIE GitHub Repo:

- https://github.com/dgoethel-noaa/2026_Sablefish_CIE/tree/main
- Contains model runs and model outputs

- CIE Doc Repo:

- <https://drive.google.com/drive/folders/1LB7eJd90IGOIImOwkEZntPfykbwmA7I7?usp=sharing>
- Contains files for review and background files

Documents For Review (available [here](#))

- **2026 Assessment of the Sablefish Stock in Alaska: 2026 Center for Independent Experts (CIE) Review Document ([here](#))** **Days 1-2**
- Appendix 3A. The Spatial Population over Regional Components (SPoRC) Model: Technical Description of the Assessment Model for Sablefish ([here](#))
- Appendix 3B. Model Building and Bridging Exercise for the Sablefish Single Region Models (Bridging from Model 25.12 to 26.1_1Reg) ([here](#)) **Day 2**
- Appendix 3C. Development of a Fleets-as-Areas (FAA) Stock Assessment Model for Alaska Sablefish (*Model 26.2 FAA*) ([here](#))
- **Appendix 3D. Results of Spatial Stock Assessment Models for Alaska Sablefish ([here](#))** **Day 3**
- **Appendix 3E. Evaluating Alternative Assessment Structures for Alaska Sablefish Under Survey Design Changes via Closed-Loop Simulation ([here](#))** **Day 2**

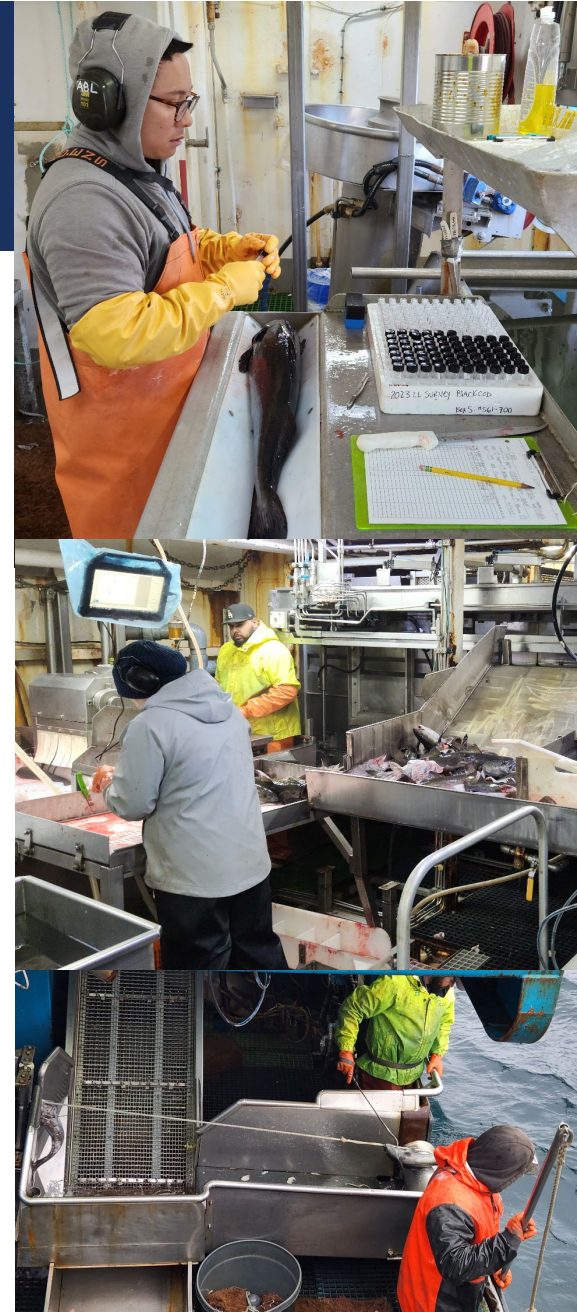


Primary Sablefish Concerns (foreshadowing)

- How best to handle recent changes in NOAA Longline Survey (LLS) design?
- Should a fleets-as-areas (FAA) model be used to avoid extrapolating LLS indices? (ToR 1+2)
- How can the spatial model be improved for potential future use as the basis of management advice? (ToR 3)

Panmictic Panacea? Demonstrating Good Practices for Developing Spatial Stock Assessments Through Application to Alaska Sablefish (*Anoplopoma fimbria*)

Matthew L. H. Cheng¹  | Craig A. Marsh² | Daniel R. Goethel²  | Peter-John F. Hulson²  | Katy Echave² | Benjamin C. Williams² | Aaron M. Berger³  | Curry J. Cunningham¹



Secondary Sablefish Concerns

- Are best practices for modeling sex- and fleet-structure being used in the assessment?
- Which data should be included in the assessment (e.g., indices, length compositions)?

Addressing complex fleet structure in fishery stock assessment models: Accounting for a rapidly developing pot fishery for Alaska sablefish (*Anoplopoma fimbria*)

Matthew L.H. Cheng^{a,*}, Daniel R. Goethel^b, Curry J. Cunningham^a

Confronting transitions in fishery fleet structure and selectivity: practical recommendations for integrated age-structured stock assessments based on simulation analysis

Matthew L.H. Cheng^a, Daniel R. Goethel^b, Peter-John F. Hulson^b, and Curry J. Cunningham^a

Misspecifying Sex-Structured Dynamics in Stock Assessment Models

Matthew L. H. Cheng¹ | Daniel R. Goethel² | Peter-John F. Hulson² | Michael J. Wilberg³ | Craig Marsh² | Curry J. Cunningham¹

Finding the signal in the noise: objective data-selection criteria improve the assessment of western Baltic spring-spawning herring

Mark R. Payne, Lotte Worsøe Clausen, and Henrik Mosegaard

Units

- Throughout the documents/presentations:
 - Mass (i.e., for spawning stock biomass, SSB, biomass, and catch) is reported in metric tons
 - Abbreviated as either mt or t (equivalent to a tonne) or kilotons (kt; equivalent to 1,000 metric tons).
 - mt is standard notation in US to avoid confusion with short ton or long ton.
 - Recruitment (age-2) and abundance are in millions of fish
 - Fishing and natural mortality are instantaneous rates (year^{-1}).



WHY DO AMERICANS HATE THE SI SYSTEM SO MUCH?

41 Action News @41actionnews · 5d

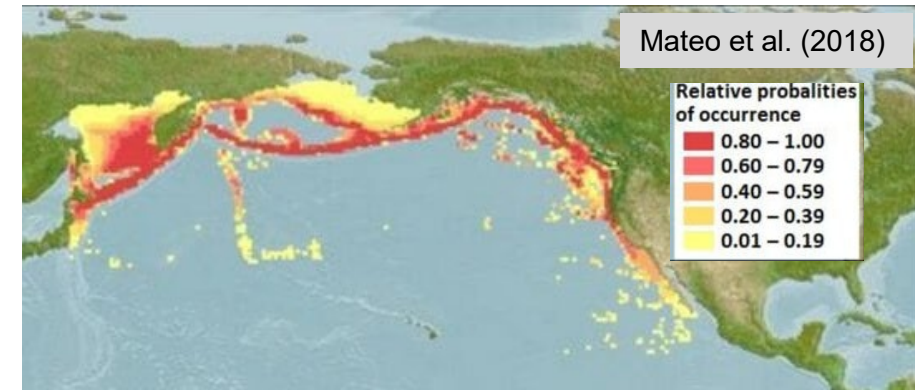
A sinkhole roughly the size of six to seven washing machines has closed the northbound lanes of State Line Road near 100th Street in Kansas City, Missouri.
bit.ly/2YAEWaj



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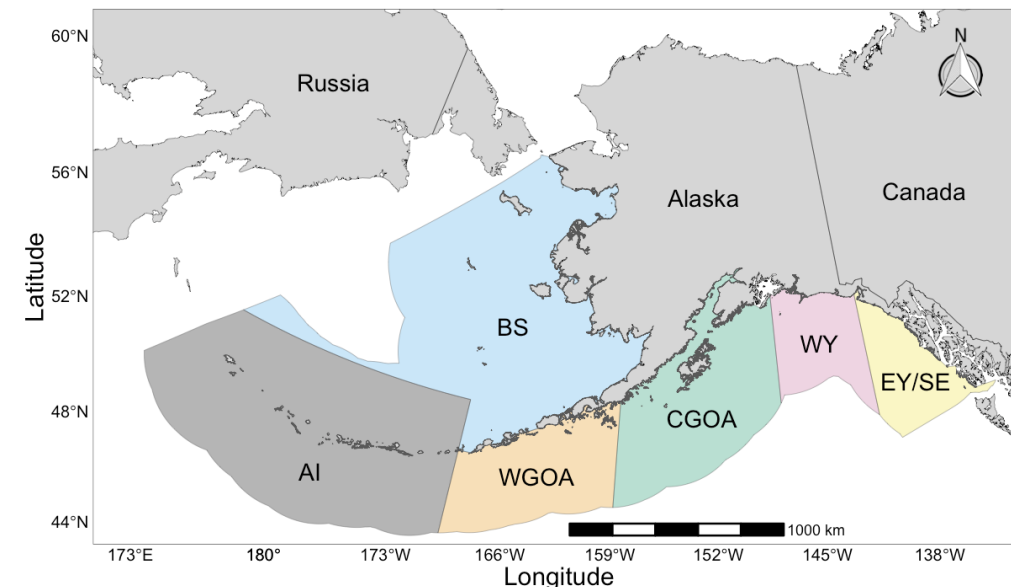
Population Structure and Spatial Distribution

- Distributed across northwestern Pacific from Mexico through Japan
 - Highest concentrations in Alaska
- No significant genetic structure across their range
 - High mobility and dispersal
- Assessed and managed as separate stocks on:
 - US West Coast (Pacific Council)
 - Canadian British Columbia (DFO)
 - **Alaska Federal Waters (North Pacific Council)**
 - Alaska State Waters (ADFG)



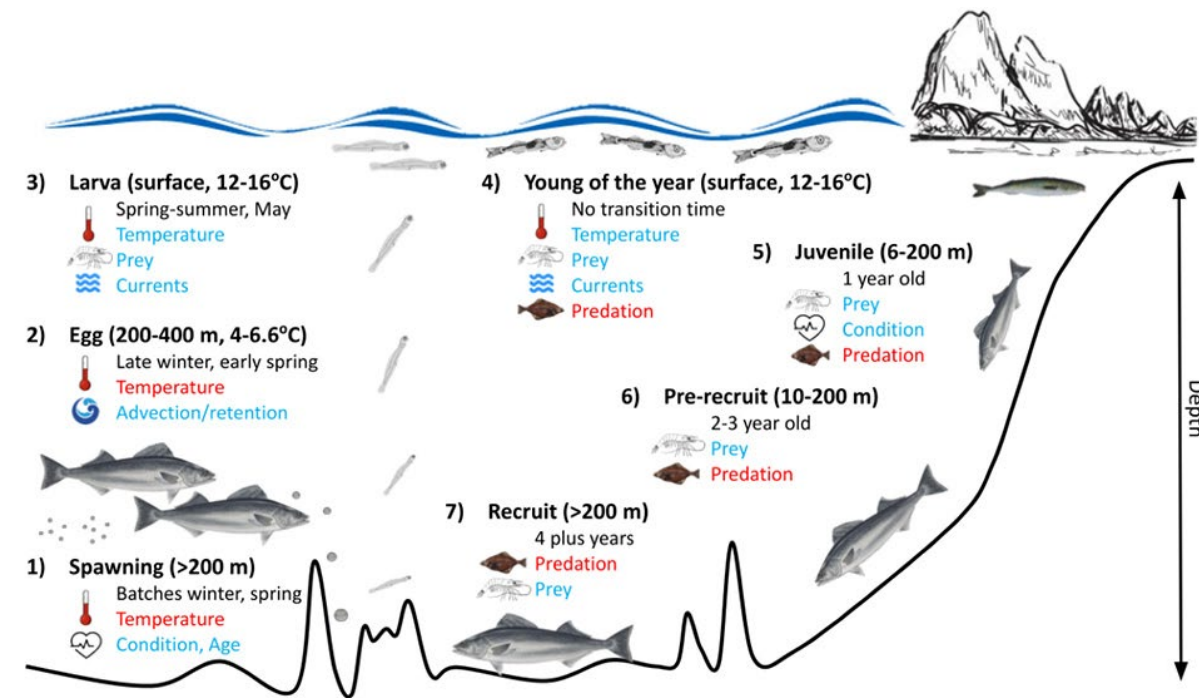
Management Structure

- One of only AK-wide managed stock for NPFMC
- Spans 3 large marine ecosystems:
 - Bering Sea (BS)
 - Aleutian Islands (AI)
 - Gulf of Alaska (GOA)
- Alaska-wide Acceptable Biological Catch (ABC) is apportioned to the 6 management regions:
 - BS, AI, WG, CG, EG (WY/SEO)
 - Based on moving 5 year average of LLS biomass proportions across regions
 - Attempts to distribute fishing pressure proportional to biomass



Life History

- Long-lived (> 90 years)
- Deep-dwelling (100 – 1500 m)
- Sexually dimorphic growth
 - Fast growing as juveniles
- Late maturation
 - $a_{50\%} \sim \text{age-7}$
 - Fully mature $\sim \text{age-12}$
- Spasmodic recruitment
- Juveniles migrate from shallow inshore to deep offshore habitat

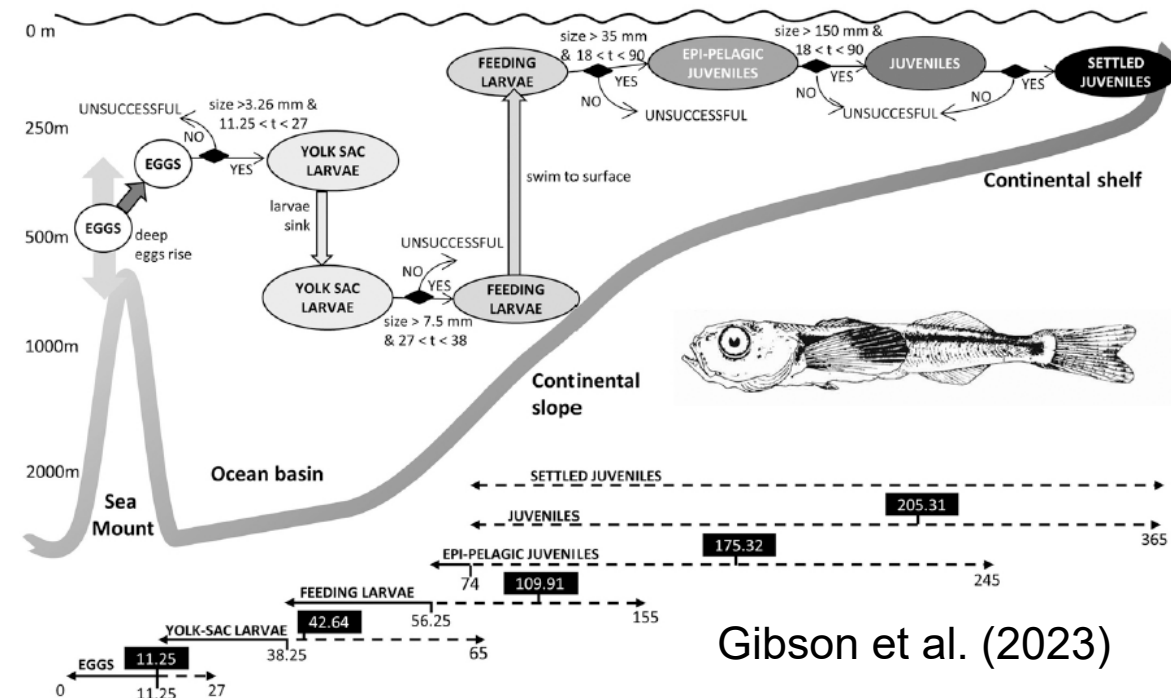
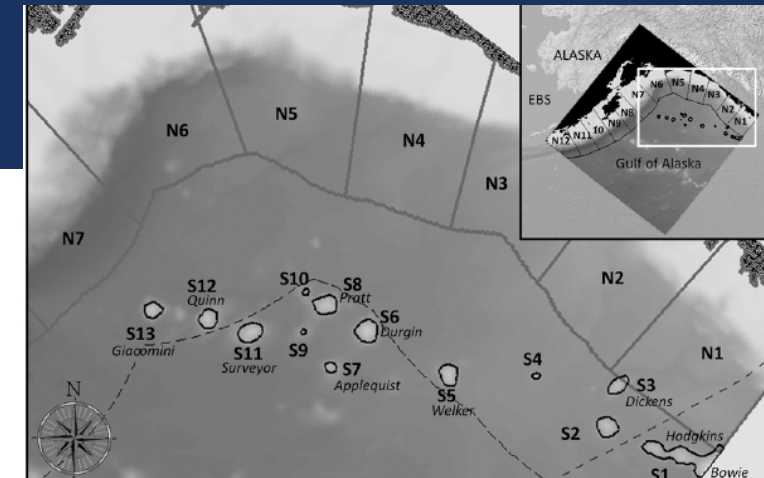


Shotwell et al. (2023)



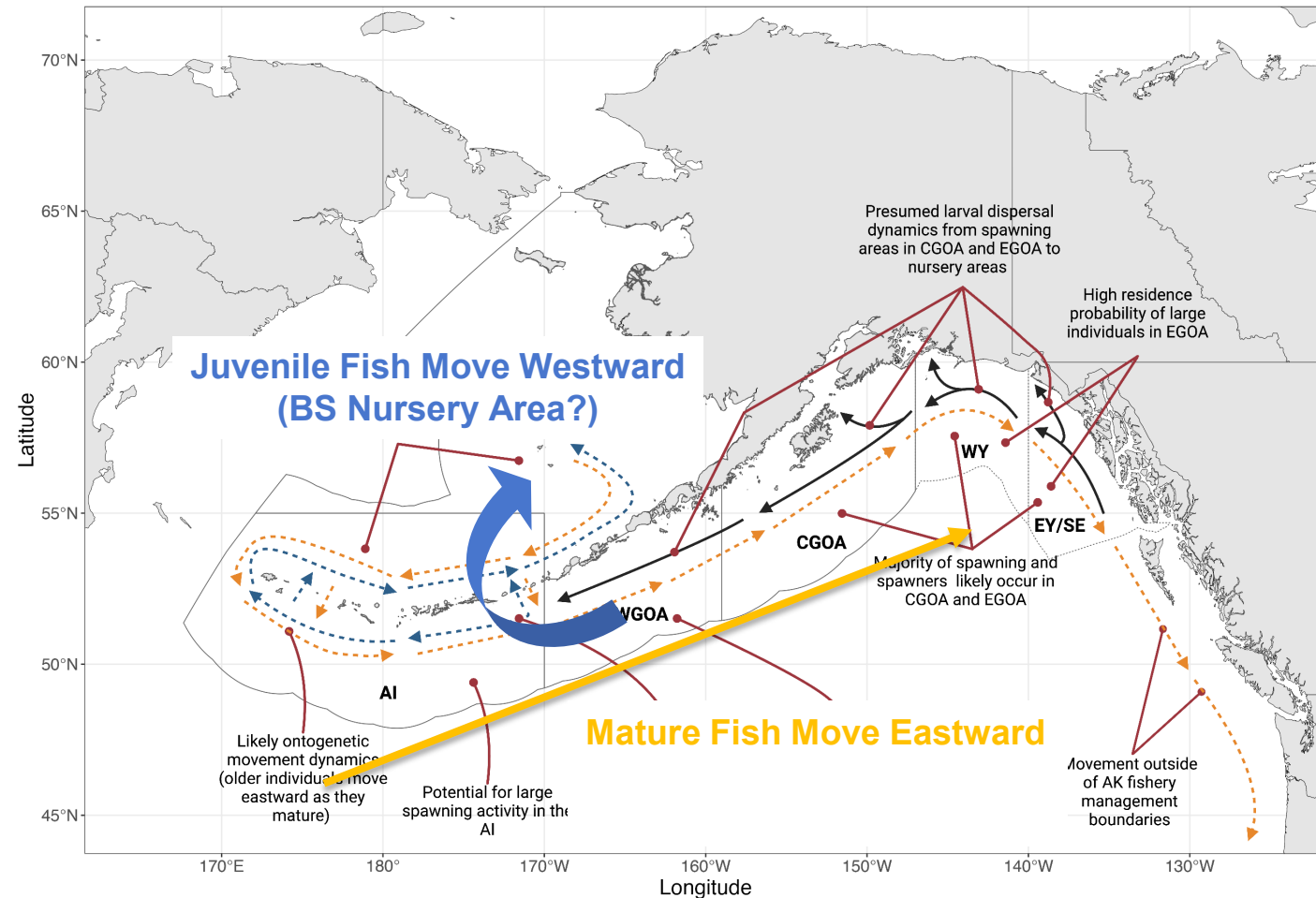
Early Life History

- Adults are found predominately on the continental slope > 200m
- Spawning occurs offshore in late winter and early spring (Jan – Mar)
 - Location of spawning is unknown but may be associated with seamounts
 - Batch spawners
 - Potential for skipped spawning especially at younger ages
- Pelagic eggs and larvae are transported to nearshore waters
- Juveniles settle on continental shelf in nearshore bays (SE AK) and eastern BS



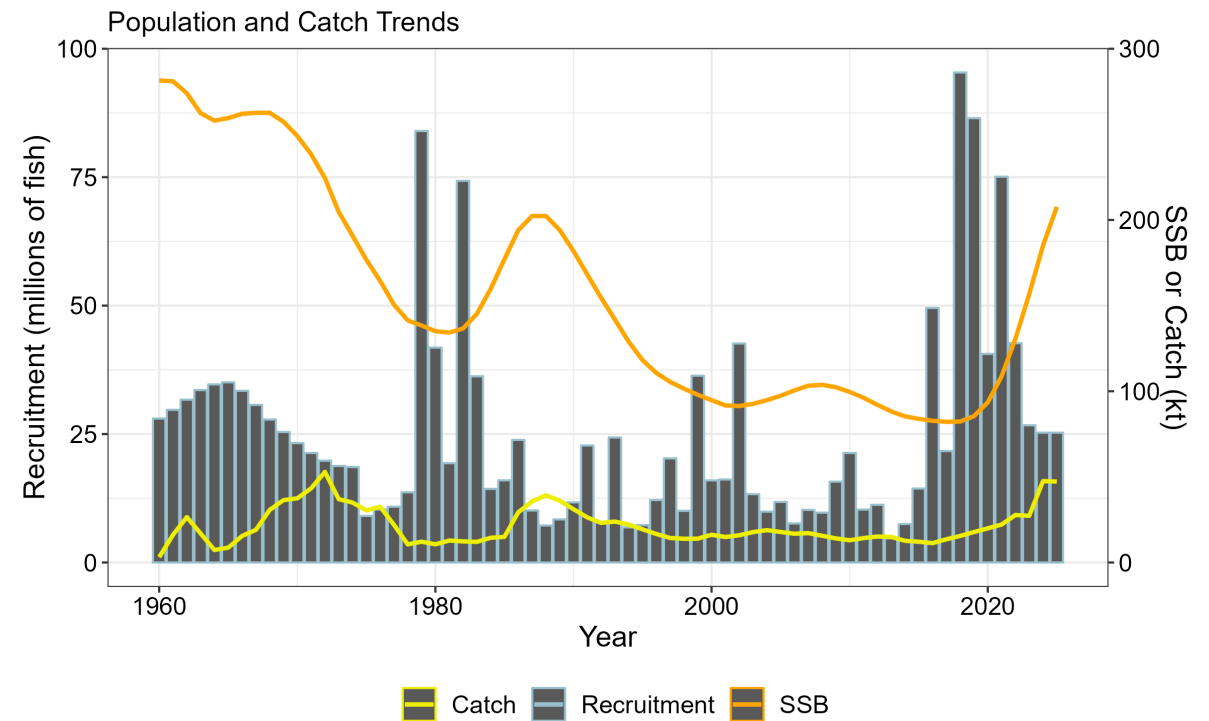
Movement and Migration

- One of the longest running mark-recapture programs in the US
- **Generally demonstrate a counter-clockwise ontogenetic movement pattern in Alaska**
- BS likely represents a historic nursery area for large cohorts
- Migrate to adult habitat (offshore slope) between ages 2 and 5
- Mature sablefish tend to be found in the CG/EG and western AI
- Extensive transboundary movement



Recruitment and Long-term Trends

- Spasmodic recruitment occurs in pulses, rapidly increasing biomass and quotas
 - Late 1970s, early 2000s, and since 2015
- Maturing fish must sustain SSB and fishery for 15+ years
- Stock was on a long-term downward trend until recently
- Recent increases coincide with onset of marine heatwaves, indicating sablefish might be a ‘climate winner’

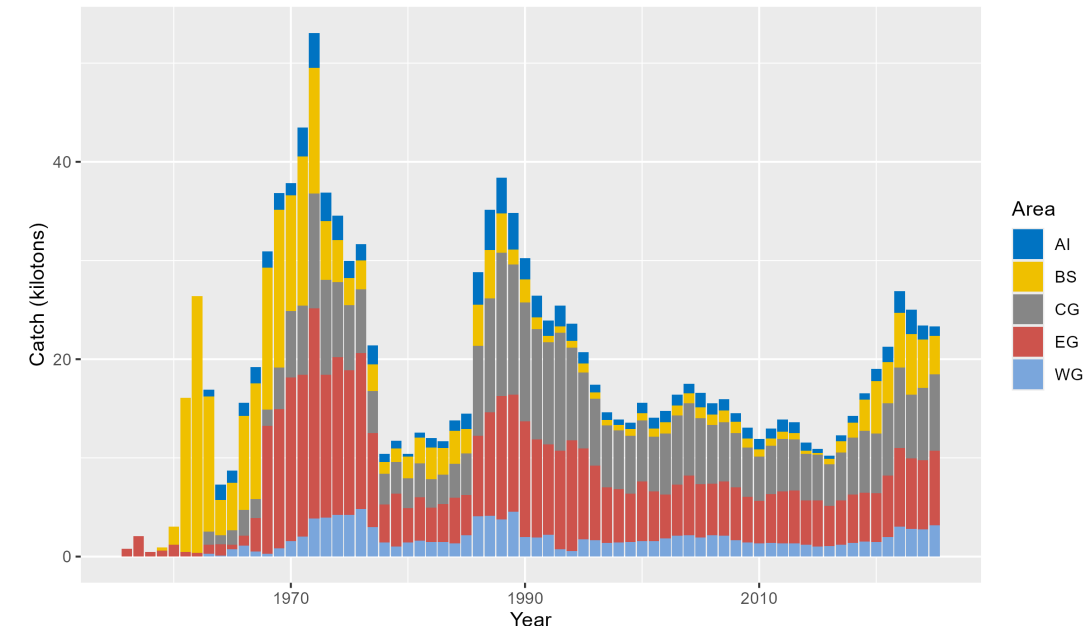


Fishery Background

- Japanese fishery started in the 1960s in BS and expanded into the GOA in 1970s
 - Fully domestic fleet by 1982
- Traditionally hook and line gear, but recent and rapid transition to longline slinky pots
 - IFQs since 1995
 - Full retention fishery
 - Primarily in GOA, with moderate BS/AI catch
- Moderate trawl catch from two fleets – pelagic trawl (BS) and bottom trawl (AI + GOA)



Catch by NPFMC Area

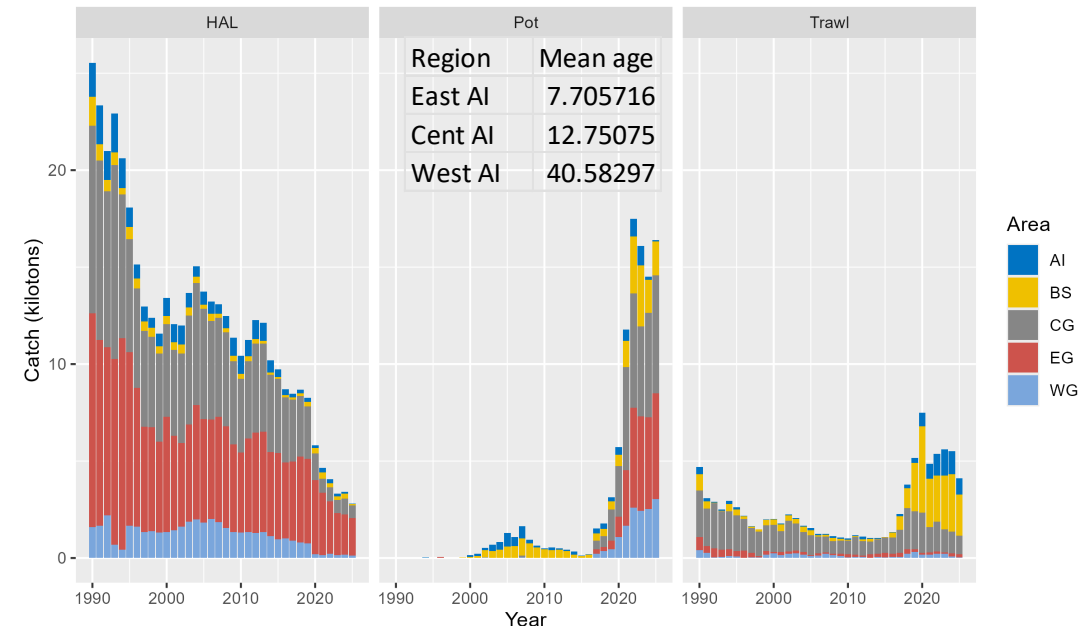


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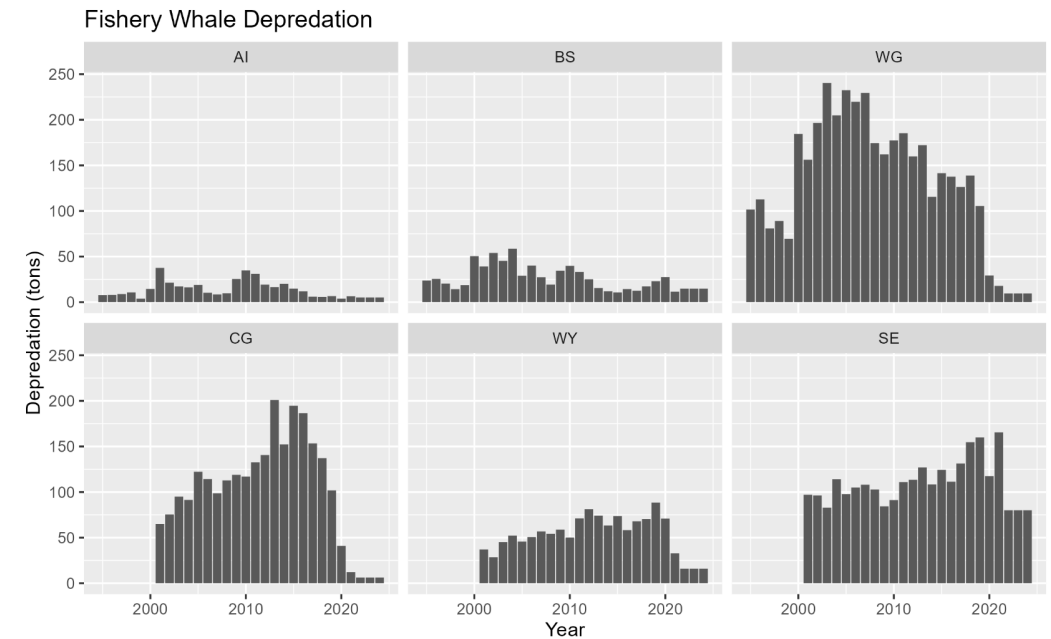


Catch by NPFMC Area



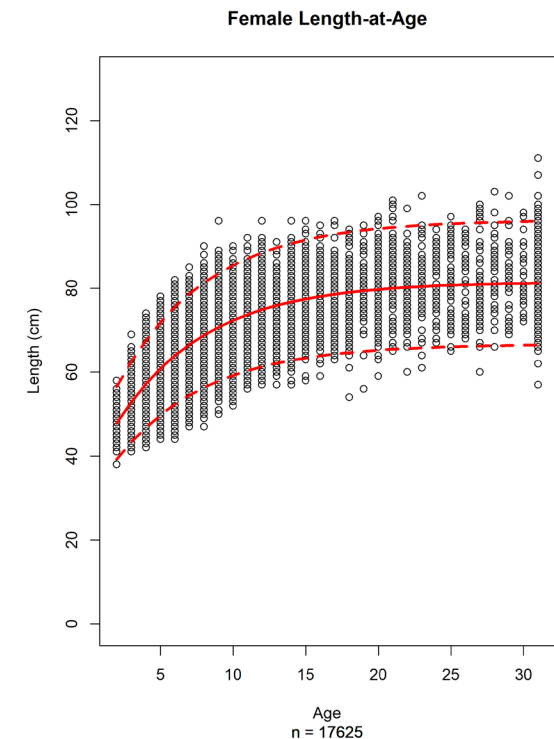
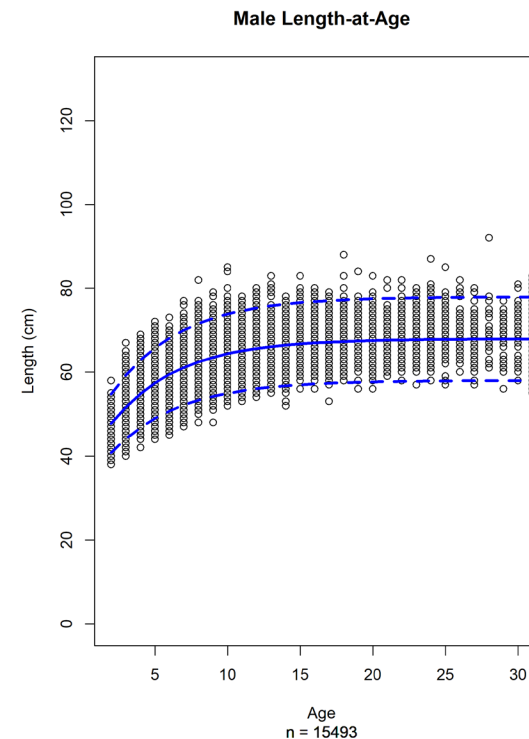
Whale Depredation

- Fishery whale depredation on hook-and-line gear had been increasing rapidly throughout the 2000s
- Primarily killer whales in the western regions (AI, BS, and WG) and sperm whales in the eastern regions (CG, EG)
- Depredation lower in AI and BS due to use of pot gear
- Since pot gear was legalized in the GOA depredation has decreased rapidly as the fishery has shifted to primarily to slinky pots



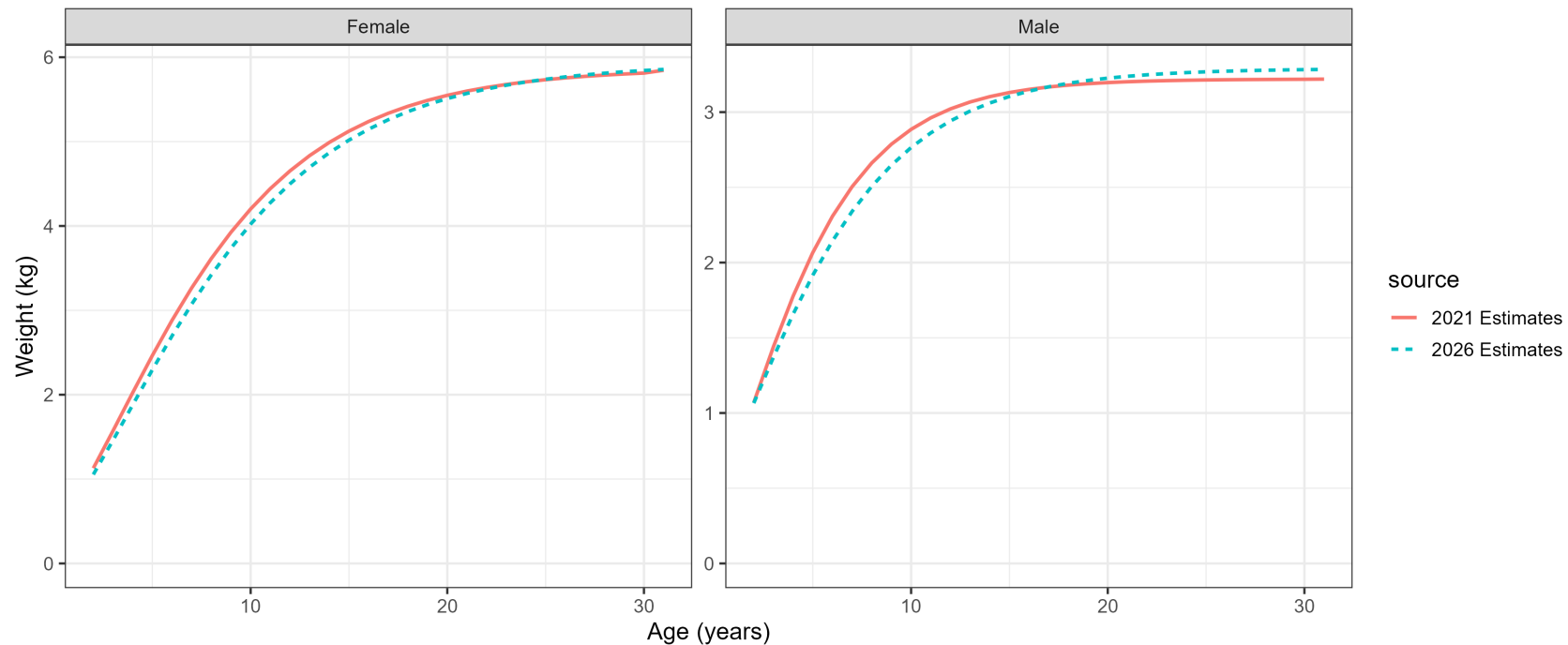
Growth

- Estimate VBGF based on paired age-lengths from LLS since 1996 (beginning of random sampling)
 - A pre-1996 growth curve is used for fitting historic Japanese LLS lengths based on bias-corrected length-stratified data (see Echave 2021)
- Strong sexual dimorphism,
 - Similar growth patterns until ~ age-5
 - Much larger asymptotic sizes for females (82 cm vs. 68 cm)
- Assessment uses size-age transition matrix to convert predicted age compositions to length compositions



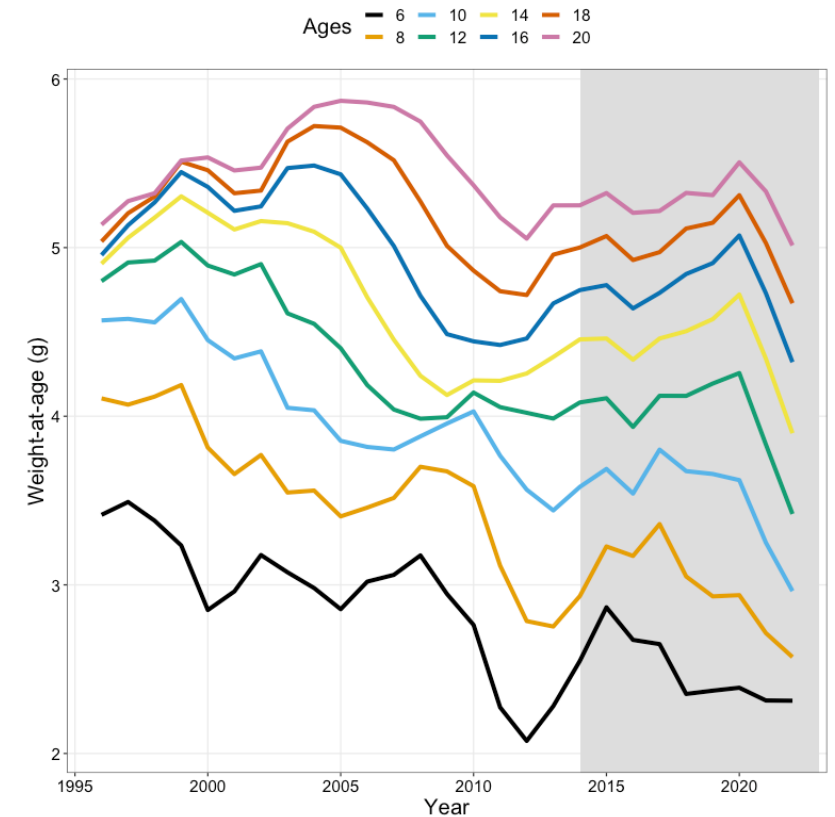
Weight

- Estimated based on paired length-weight observations from LLS since 1996 (beginning of weight sampling)
 - A single weight-at-age curve is used in the assessment



Density-Dependence(?)

- Reduced length- and weight-at-age coincides with recent high recruitment events
 - Likely coupled with environmental drivers
- Time-varying growth could complicate the use of length compositions in the assessment if not adequately modeled
- Time-varying weight could have implications for BRPs, SSB, and catch advice

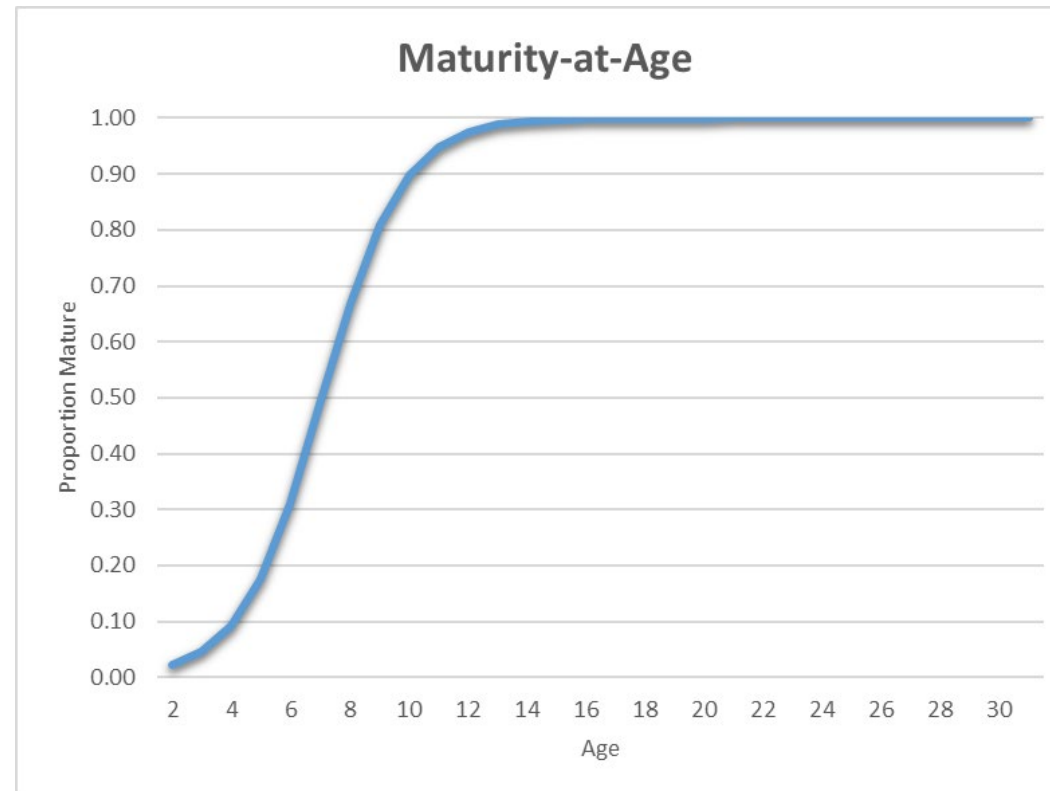


“Slim pickings?”: Extreme large recruitment events may induce density-dependent reductions in growth for Alaska sablefish (*Anoplopoma fimbria*) with implications for stock assessment

Matthew L.H. Cheng^a, Daniel R. Goethel^b, Peter-John F. Hulson^b, Katy B. Echave^b, and Curry J. Cunningham^a

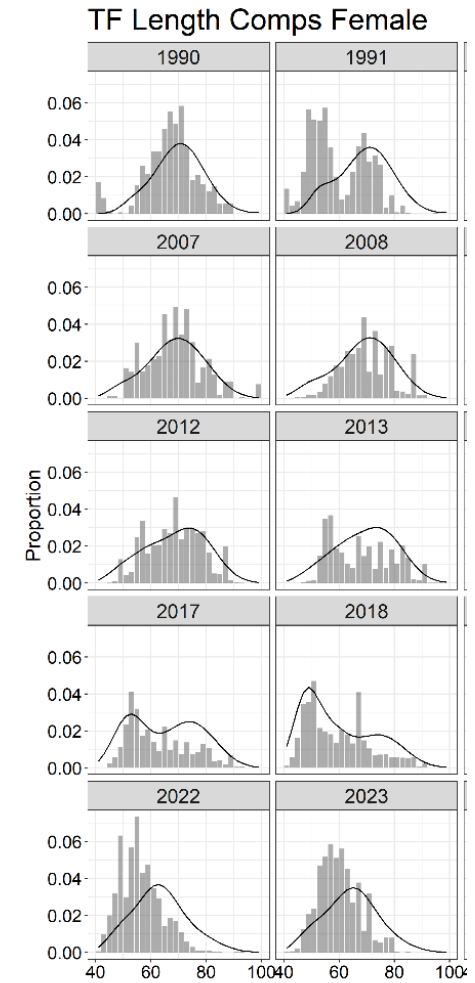
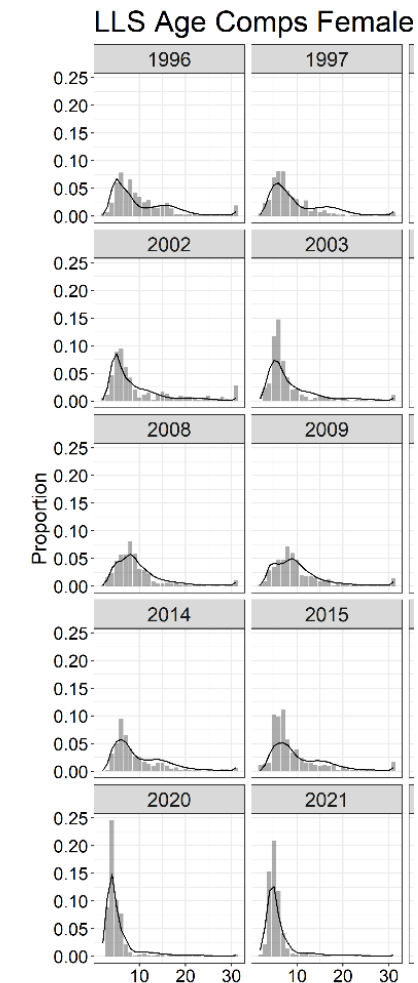
Maturity

- One maturity time block based on histological samples
 - $a_{50\%} \sim \text{age-7}$
 - Fully mature $\sim \text{age-12}$



Age at Recruitment and Sex Ratio

- Sablefish appear on continental slope around age-2 at fork length of 45 cm
- First appear in LLS and LLF around the same age/size, but in relatively low frequency
- Higher interactions with pelagic trawl fishery in the EBS at smaller sizes
- Assessment models age-2+, length 40 cm+
- No indication of a strong divergence from 50:50 sex ratio based on LLS sampling



Ageing Error

- Sablefish are difficult to age
- Ageing error matrix developed based on 172 known-age otoliths from age-1 tagging studies
- The ageing error matrix is directly incorporated into the assessment predictions of age compositions to account for uncertainty in the ageing process.

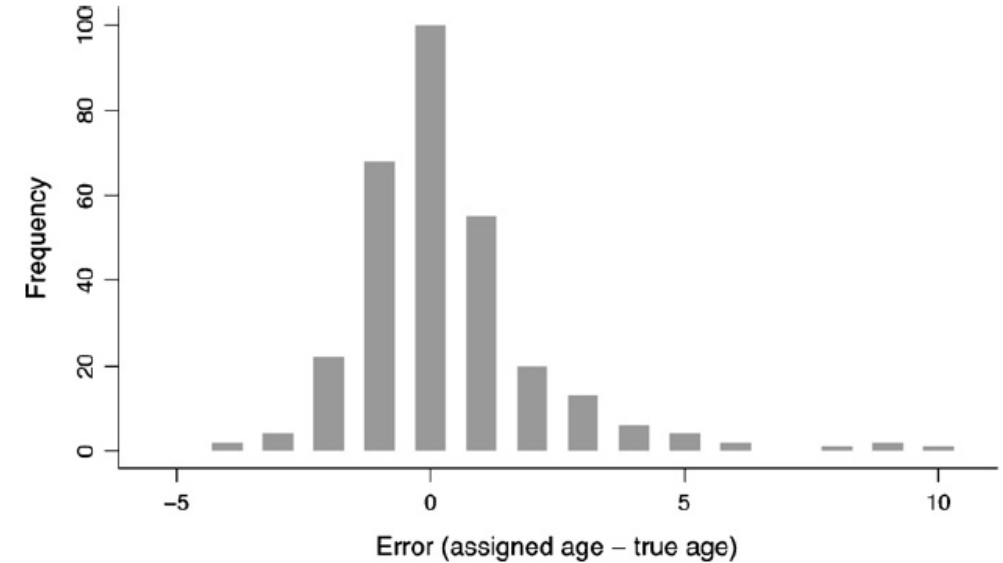
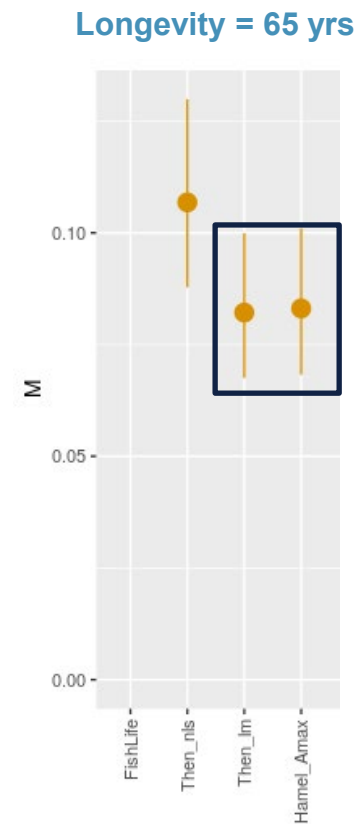


Fig. 4. Distribution of age reading errors for sablefish of ages 5–8 pooled. There are three readings of each otolith in the counts.



Natural Mortality

- Natural mortality is difficult to estimate in the assessment
 - $a_{max} \sim 94$ years in AK (~ 102 for WC)
 - Only 6 fish > 70 years old in LLS age compositions
- M estimated to be 0.085 from tagging data
- An average longevity of 65 years equates to $M \sim 0.085$
- WC and BC assessments estimate $M \sim 0.09$.
- Current assessment M prior is $\sim N(\ln(0.085), 0.1^2)$



Key Considerations for Assessment

- Long-lived (low M), difficult to age (aging error matrix integrated), and relatively late maturation
- Widely distributed (AK-wide assessment) but no population structure (single population)
- High mobility with ontogenetic migrations (differential availability to fishery/surveys across regions)
- Spasmodic and cyclic recruitment, likely influenced by environmental factors (lack of a well-defined SRR)
- Sexually dimorphic (sex-specific growth/selectivity)
- Multiple fisheries with different interactions with sablefish (multiple fleets with different selectivity)
 - Rapid changes in fixed gear fishery (time-varying selectivity)
 - Depredation has been high but decreased due to pot gear usage



